



Branch/Semester	B.Tech CSE (AI & ML/DS) / Semester II
Subject Name:	Object Oriented Design and Programming
Subject Code:	01AI0103
Assignment:	Practice Questions (OODP)
Date:	20 th March 2026
Faculty Name:	Prof. Abhishek Chauhan

Sr.	Questions (Introduction to OOP)	CO	BL
1	Explain the differences between Procedural Programming and Object-Oriented Programming. Write a C++ program to demonstrate both approaches.	CO1	BL2
2	Describe the basic structure of a C++ program. Write a program using header files, namespaces, and preprocessors.	CO2	BL3
3	Explain all OOP principles (Encapsulation, Abstraction, Inheritance, Polymorphism) with suitable real-world examples.	CO1	BL2
4	Write a C++ program to demonstrate the use of access specifiers (public, private, protected) in a class.	CO2	BL3
5	Analyze the advantages of OOP over procedural programming in terms of reusability and maintainability.	CO3	BL4
6	Develop a simple C++ program using library functions and explain each component of the program.	CO2	BL3
7	Compare structure and class in C++ with a suitable program and explain their differences.	CO1	BL4

Sr.	Questions (Programming Basics)	CO	BL
1	Write a C++ program to accept user input and display formatted output using manipulators like setw.	CO2	BL3
2	Demonstrate the use of bool data type and conditional statements with a suitable example.	CO2	BL3
3	Write a program to perform type conversion (implicit and explicit) and explain the results.	CO2	BL3

4	Develop a program to display a formatted table using various output manipulators.	CO2	BL3
5	Analyze the importance of input/output streams in C++ programming with examples.	CO3	BL4
6	Write a program to handle different data types and demonstrate type casting.	CO2	BL3
7	Identify and correct errors in a given program related to input/output operations.	CO3	BL4

Sr.	Questions (Functions)	CO	BL
1	Write a program to swap two numbers using both call by value and call by reference. Compare the results.	CO2	BL3
2	Explain function overloading with an example program demonstrating multiple functions with same name.	CO2	BL3
3	Write a program to demonstrate default arguments and explain their advantages.	CO2	BL3
4	Compare macro functions and inline functions with suitable examples and performance analysis.	CO3	BL4
5	Write a program using friend function to access private data of a class.	CO2	BL3
6	Develop a function to calculate power (m^n) using default arguments and explain its working.	CO2	BL3
7	Analyze the role of functions in modular programming and code reusability.	CO3	BL4

Sr.	Questions (Objects and Classes)	CO	BL
1	Define a class and object. Write a program to demonstrate constructors and destructors.	CO4	BL3
2	Explain different types of constructors with suitable examples in C++.	CO1	BL2
3	Write a program to create a class for a bank account system with deposit and withdrawal functions.	CO4	BL6
4	Demonstrate how class members are accessed using objects with a suitable example.	CO2	BL3
5	Analyze the importance of destructors in memory management with example.	CO3	BL4
6	Compare structures and classes with respect to data hiding and security.	CO1	BL4
7	Design a class to represent complex numbers and perform addition using constructors.	CO4	BL3

Sr.	Questions (Operator Overloading)	CO	BL
1	Explain operator overloading. Write a program to overload unary operator (++) for a class.	CO2	BL3
2	Write a program to overload binary operator (+) for adding two complex numbers.	CO4	BL3
3	Demonstrate type conversion using operator overloading with a suitable example.	CO2	BL3
4	Analyze the limitations and pitfalls of operator overloading in C++.	CO3	BL4
5	Write a program to overload comparison operator (==) for string comparison.	CO2	BL3
6	Develop a class FLOAT and overload arithmetic operators (+, -, *, /).	CO4	BL3
7	Compare operator overloading with function overloading in terms of flexibility and usage.	CO3	BL4

Sr.	Questions (Inheritance)	CO	BL
1	Explain the concept of inheritance. Write a program to demonstrate single inheritance.	CO4	BL3
2	Differentiate between public, private, and protected inheritance with examples.	CO1	BL2
3	Write a program to demonstrate multilevel inheritance with proper class hierarchy.	CO4	BL3
4	Analyze how constructors are called in inheritance with suitable example.	CO3	BL3
5	Design a class hierarchy for an employee management system using inheritance.	CO4	BL6
6	Compare base class and derived class with suitable examples.	CO1	BL2
7	Identify and debug common errors in inheritance-based programs.	CO3	BL4

Sr.	Questions (Polymorphism)	CO	BL
1	Explain runtime polymorphism. Write a program using virtual functions.	CO4	BL3
2	Demonstrate function overriding with a suitable example.	CO2	BL3
3	Explain the concept of this pointer with example.	CO1	BL2
4	Write a program using pure virtual functions and abstract class.	CO4	BL3
5	Analyze the difference between compile-time and runtime polymorphism.	CO3	BL4
6	Develop a program demonstrating runtime polymorphism using base class media and derived classes.	CO4	BL3
7	Debug errors in polymorphism-related programs and explain the issues.	CO3	BL4

Sr.	Questions (Streams and Files)	CO	BL
1	Write a program to read and write data from a text file using file streams.	CO2	BL3
2	Develop a program to copy contents from one file to another and explain file handling steps.	CO2	BL3
3	Demonstrate formatted and unformatted input/output operations in C++.	CO2	BL3
4	Analyze different file modes and their significance in file handling.	CO3	BL4
5	Design a phonebook application using file handling techniques.	CO4	BL6
6	Write a program to handle binary files and perform read/write operations.	CO2	BL3
7	Identify and debug common file handling errors in C++ programs.	CO3	BL4

Sr.	Questions (Templates & Exception Handling)	CO	BL
1	Explain function templates and write a program to find the minimum value in an array.	CO2	BL3
2	Develop a class template for vector operations and demonstrate its usage.	CO2	BL6
3	Write a program to demonstrate exception handling using try, catch, and throw.	CO2	BL3
4	Explain multiple catch blocks and write a program to handle different exceptions.	CO3	BL3
5	Analyze the role of Standard Template Library (STL) in efficient programming.	CO4	BL4
6	Demonstrate inheritance with templates using a suitable program.	CO2	BL3
7	Identify and debug errors in exception handling programs and suggest improvements.	CO3	BL4